Worksheet # 3 Part B Upload solutions
Calculus Physics II

Name: Key Date: _____

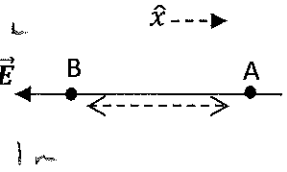
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Upload the solution to questions 11 through 15 below

11) An object with a net negative charge of 200 μC moves in an electric field $\vec{E}$ that changes along the x axis as $\vec{E} = [(6\,N/Cm^2)x^2]\hat{x}$ from point B at 1 m to point A at 2 m. Determine the work the field does on the charge moving from point B to point A?

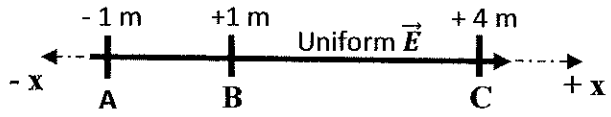
$W = \int \vec{F}\cdot d\vec{x} = q\int \vec{E}\,d\vec{x} = q\int |\vec{E}|\,dx$

$W = q\int_B^A (6\,N/Cm^2)\,x^2\,dx = q\,\frac{1}{3}(6\,N/Cm^2)\,x^3\Big|_1^2$



$W = 0.032\,J - 4\times10^{-4}\,J = 0.0028\,J = 2.8\times10^{-3}\,J$

12) A uniform electric field is shown below with positions A, B and C.



A) Determine the electric field strength if a particle has a net positive charge of 100 μC and a potential energy difference for the particle between points A and C of $\Delta PE_{C\text{ to }A} = 0.5$ J.

$\Delta PE = -q|\vec{E}|\Delta x$ | Solve For $|\vec{E}|$ | $|\vec{E}| = \frac{\Delta PE}{q\Delta x} = 1000\,V/m$

B) Determine the electric potential difference between B and C.

$\Delta V_{BC} = |\vec{E}|\,\Delta x_{BC} = 3000V$

C) If position A is at an electric potential of + 2,000 V, determine the electric potential at point C.

$\Delta V_{AB} = |E|\,\Delta x_{AB} = 2000V$

$V_B = V_A - \Delta V_{AB} = 0V$ | $V_C = V_B - \Delta V_{BC}$

$V_C = -3000V$

13) The electric potential is changing along the x-axis given by the equation below

$$V(x) = (3\,V/m^3)x^3 - (18\,V/m^2)x^2 + (20\,V/m)x + 2V$$

A) Determine the position(s) along the x axis where the magnitude of the electric field is zero.

$x_1 = 0.67\,m$

$x_2 = 3.0\,m$

See Attached

B) Determine the potential difference between the zero electric field points.

$\Delta V = -28.4\,V$

See Attached

14) A thin wire of radius R with a uniform linear charge density of λ is shown below, determine the electric potential at point P at a distance of "s" above the center of the wire.

$$V = K\int \frac{dq}{r}$$

$$V = K\int \frac{\lambda R\, d\theta_R}{r}$$

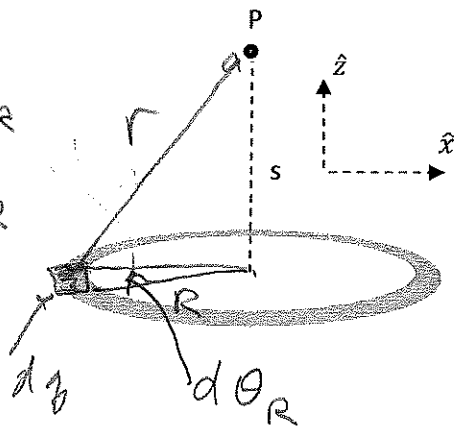
$$V = \frac{K\lambda R}{(s^2+R^2)^{1/2}}\int_0^{\pi} d\theta_R$$

$$dq = \lambda\, dL$$

$$dL = R\, d\theta_R$$

$$dq = \lambda R\, d\theta_R$$

$$r^2 = s^2 + R^2$$



$$V = \frac{2\pi\lambda R K}{(s^2+R^2)^{1/2}}$$

15) An electric motor is connected across an ac voltage source given by the equation below and the rms current that the motor draws is $I_{rms} = 4$ A.

$$V(t) = (325\text{ V})\sin[(377/s)t].$$

A) Determine the resistance of the motor if only the power source and motor are wired in the series.

$V_p = 325\text{ V}$ | $V_{rms} = \frac{V_p}{\sqrt{2}} = 230\text{V}$

B) Determine the average power used by the motor.

$$V_{rms} = I_{rms} R$$

$$R = \frac{V_{rms}}{I_{rms}} = 57.5\,\Omega$$

$$P_{avg} = I_{rms} V_{rms}$$

$$P_{avg} = (4\text{ A})(230\text{V})$$

$$P_{avg} = 920\text{W}$$

13) $\vec{E} = -\left(\frac{dV}{dx}\right)\hat{x}$ | $|\vec{E}| = E_x = -\frac{dV}{dx}$

A) $E_x = \frac{d}{dx}\left[3x^3 - 18x^2 + 20x + 2\right] = 0$

$E_x = 9x^2 - 36x + 20 = 0$

using quadratic equation

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $x_1 = 0.67\text{m}$

$x_2 = 3.3\text{m}$

B)

$V(0.67\text{m}) = 3(0.67\text{m})^3 - (18)(.67)^2 + 20(0.67) + 2$

$= 8.2\text{V}$

$V(3.3\text{m}) = 3(3.3\text{m})^3 - (18)(3.3)^2 + 20(3.3) + 2$

$= -20.2\text{V}$

$\Delta V = V_2 - V_1 = -28.4\text{V}$